

Table 1: In-context learning datasets in our experiments.

Dataset	# Demos	Modality	Size
ARC-AGI (Chollet, 2019)	2~10	Symbolic	835
MiniARC (Kim et al., 2022)	2~8	Symbolic	149
1D-ARC (Xu et al., 2024)	3	Symbolic	901
SCAN (Lake & Baroni, 2018)	5~8	Textual	1,000
MiniSCAN (Nye et al., 2020)	14	Textual	1,000
COGS (Kim & Linzen, 2020)	10	Textual	1,000
SALT (Zheng et al., 2025a)	4	Textual	1,200
List Function (Rule, 2020)	3	Numerical	1,250
RAVEN (Zhang et al., 2019)	2	Numerical / Symbolic	1,259
Total			8,594

Chain-of-Thought Prompting In contrast, Chain-of-Thought Prompting involves a two-stage response process. First, the LLM generates explicit intermediate reasoning or rationale conditioned on the instructions, demonstrations $\mathcal{D}$, and test input x_{test} . Second, it produces the final output y_{test} based on this rationale, alongside the original context (instructions, $\mathcal{D}$, and x_{test}). This process is formally expressed as:

$$p(\text{rationale} \mid x_{\text{test}}, \mathcal{D}, \text{Instructions}) \cdot p(y_{\text{test}} \mid \text{rationale}, x_{\text{test}}, \mathcal{D}, \text{Instructions})$$

Notably, the demonstration examples $\mathcal{D}$ are identical in both paradigms and do not include explicit reasoning steps. Consequently, our targeted task formulation differs from the few-shot CoT approaches commonly employed in standard QA tasks, where demonstrations explicitly illustrating CoT reasoning steps are provided. Additionally, we experimented with advanced reasoning frameworks, including ReAct and Tree-of-Thought prompting, in which explicit reasoning guidance is provided prior to the task instructions. The detailed prompting template is presented in Appendix D.

3 Datasets and models

Datasets We conduct experiments on a diverse selection of pattern-based in-context learning datasets spanning multiple modalities¹: 1) **Symbolic**: Pattern-based transformations between symbolic matrices, e.g., ARC-AGI and MiniARC. 2) **Textual**: Rule-based translations between natural language and artificial languages, e.g., SCAN and COGS. 3) **Numerical**: Pattern-based or function-based projections between numerical vectors or matrices, e.g., List Functions and RAVEN.

All selected datasets are reasoning-intensive, stress-testing the abstract and inductive reasoning abilities of LLMs. Details of datasets are provided in Table 1. We include further data processing details in Appendix C.

Models We evaluated 16 open-source and proprietary LLMs with varying parameter sizes, with details in Appendix A. Note that long-CoT reasoning models, such as o1 and Deepseek-R1, are analyzed separately from the main group of LLMs due to their limited compatibility with direct answering. As these models are specifically optimized for multi-step explicit reasoning, they provide a crucial test case for our hypothesis and are therefore given a dedicated analysis in Section 6.

4 Main results

The main experimental results are illustrated in Figure 2 (full results in Appendix E). Across nine ICL benchmarks, LLMs employing **direct answering substantially outperform CoT**, achieving a relative improvement of **20.42%** (absolute 5.10%). Compared to ReAct and ToT, direct answering yields relative improvements of **36.34%** and **47.17%** (absolute 8.02% and 9.64%), respectively. In terms of task modality,

¹<https://github.com/HKUST-KnowComp/CoT-ICL-Eval>



Figure 2: Detailed benchmark performance of LLMs with direct answering, CoT, ReAct, and ToT. Gemma2 models were excluded from ARC-AGI experiments due to limited context length.

the performance gap between direct answering and CoT is **most significant on symbolic ICL tasks** (i.e., ARC-AGI, MiniARC, 1D-ARC, and RAVEN), with a relative improvement of 41.88%; in contrast, this advantage decreases to 10.42% on textual ICL tasks (i.e., SCAN, MiniSCAN, COGS, and SALT).

Regarding model size, since most benchmarks used in our study can be considered relatively out-of-distribution

compared to the LLM training corpora², smaller LLMs (e.g., Llama3.1-8B, Qwen2.5-7B) tend to exhibit lower overall performance, as well as more pronounced limitations when utilizing CoT or other reasoning variants. In contrast, larger models (e.g., GPT-4o, Deepseek-V3) achieved better overall performances, in which reasoning frameworks occasionally achieve performance comparable to direct answering. Nevertheless, there are only a few entries in which reasoning frameworks yield a positive outcome from the additional consumption of inference tokens.

Moreover, for two benchmarks that allow flexibility in the number of demonstrations (COGS and List Function), we conduct experiments by varying the demonstration count in the context, ranging from 3 to 15. As illustrated in Figure 1b, the performance gap between direct answering and CoT **widens as the number of shots increases**. This further substantiates the limitations of CoT under different contextual configurations.

These experimental findings reveal a surprising "curse" of CoT, where reasoning frameworks consistently underperform direct answering in pattern-based ICL tasks—with more sophisticated variants (ReAct, ToT) performing even worse. This counterintuitive phenomenon challenges conventional assumptions about the benefits of explicit reasoning in LLMs and motivates our systematic investigation into the underlying mechanisms behind this performance degradation.

5 Why chain-of-thought fails in in-context learning?

In this section, we systematically diagnose the root causes of CoT’s inefficacy through a hypothesize-and-test methodology. We design targeted experiments to validate or refute potential explanations for this limitation. Details of all four experiments are in Appendix B.

5.1 The Contextual Distance Curse: how CoT disrupts few-shot learning

In-context learning, as delineated by Brown et al. (2020), assumes that few-shot demonstrations are presented as a coherent, uninterrupted sequence, enabling the model to process them as a unified contextual signal for learning. However, under Chain-of-Thought prompting, the insertion of intermediate rationales between demonstrations and the final answer prediction may disrupt this continuity. We thus propose our first hypothesis:

Hypothesis 1. The CoT rationale increases the contextual distance between demonstrations and answers, disrupting the few-shot learning structure and thereby degrading performance.

Formally, we define *contextual distance* as the number of tokens separating the end of the in-context demonstrations (i.e., the input-output pairs provided in the prompt) from the position in the sequence where the model begins generating the final answer output. In the context of Chain-of-Thought prompting, this distance generally equals the token length of the generated reasoning rationale, which is inserted between the demonstrations and the answer.

To test this hypothesis, we designed two controlled experiments to isolate and evaluate the effect of contextual distance and CoT:

Dummy Rationale Experiment To disentangle the semantic content of CoT from its structural impact, we instructed LLMs to generate a semantically neutral "dummy" rationale prior to predicting the final answer, thereby preserving the contextual distance while eliminating reasoning-specific effects. We controlled two variables: modality and length. For modality, we considered textual and symbolic data. In the textual condition, LLMs recited excerpts from Shakespeare’s Sonnets; in the symbolic condition, they generated a countdown list from a specified integer to one. These tasks were chosen to minimize generation variance and prevent unbounded outputs. For length, we varied the dummy rationale size: reciting 1, 2, 4, or 8 sonnets (approximately 150 tokens per sonnet) and counting down from 50, 100, 200, or 400 (approximately 3 tokens per number). This yielded contextual distances ranging from 150 to 1200 tokens, encompassing typical CoT rationale lengths (150 to 500 tokens).

²The SCAN dataset might be subject to data contamination to some extent.